

CONSISTENCY WITHOUT COHERENCE

On the Unfalsifiable Substrate of Empirical Physics

Eris Arete

Independent Researcher

Abstract

Physics presents itself as the paradigm of empirical accountability, yet its mathematical apparatus is judged by internal consistency rather than empirical coherence. This essay argues that the asymmetry—between a substantive theory layer subject to empirical test and a mathematical apparatus layer insulated by consistency—systematically favors framework preservation through auxiliary elaboration over framework revision. The pattern is illustrated through five cases: the luminiferous aether, dark matter and dark energy, the cosmological constant problem, the proliferation of competing quantum gravity programs, and the symmetrization postulate. Each exhibits the same structural dynamic: framework commitments and the empirical record diverge; the substantive theory is elaborated; the framework is preserved; the cumulative effect is the accumulation of unfalsifiable elaborations and the foreclosure of alternatives at coherence-based grounds. The essay specifies the criteria—Lakatosian progressiveness applied at framework level, combined with explicit ansatz commitment—by which alternative frameworks should be evaluated, addressing the circularity worry that the call to alternatives might otherwise raise. The essay concludes that the philosophy of physics must treat alternative-framework construction as a structural responsibility rather than a marginal pursuit: only through the development of formally precise alternatives can foundational commitments be rendered accountable to evidence at the level where the discipline currently exempts them.

Keywords: consistency vs. coherence; mathematical apparatus; framework preservation; dark matter; cosmological constant; quantum gravity; symmetrization postulate; relational ontology; alternative frameworks; philosophy of physics.

I. The Standard View

Physics presents itself, both to philosophers of science and to the broader public, as the paradigm of empirical accountability. Its claims are formulated such that they could in principle be wrong; its theories are revised in the face of disconfirming evidence; its progress consists in the replacement of less accurate accounts by more accurate ones. The standard philosophical accounts of how physics achieves these features form a familiar lineage. Popper's demarcation criterion holds that scientific theories distinguish themselves from non-scientific systems by being framed in ways that admit of empirical falsification; a claim that cannot in principle be refuted by any conceivable observation is, on Popper's account, not a scientific claim at all.¹ Lakatos refined this picture by recognizing that working scientists rarely abandon a theory on the basis of a single disconfirming result; they operate within research programmes whose hard cores are protected by belts of auxiliary hypotheses that absorb anomalies, with the programme as a whole adjudicated by whether it remains progressive—predicting novel facts—or degenerates into a series of ad hoc rescues.² Kuhn supplied the historical and sociological complement, describing the periods of normal science in which a paradigm's puzzle-solving capacity is taken for granted, and the periods of revolutionary science in which an accumulation of anomalies prompts the paradigm's replacement.³

These accounts have been developed, criticized, and refined across the subsequent decades. Contemporary philosophers of physics have offered substantial defenses of the empirical-accountability picture against various challenges. Maudlin has argued that the foundational debates in quantum mechanics and relativity have been settled, in their philosophical essentials, by careful attention to what the theories themselves say about the world.⁴ Earman has developed extended analyses of how general relativity is and is not subject to empirical revision, distinguishing the

¹K. R. Popper, *The Logic of Scientific Discovery* (Routledge, London, 1959). The demarcation criterion is developed throughout but receives its canonical statement in Chapter 4. Subsequent refinements by Popper himself in later works do not affect the substance of the criterion as it is invoked here.

²I. Lakatos, "Falsification and the Methodology of Scientific Research Programmes," in *Criticism and the Growth of Knowledge*, edited by I. Lakatos and A. Musgrave (Cambridge University Press, Cambridge, 1970), pp. 91–196.

³T. S. Kuhn, *The Structure of Scientific Revolutions* (University of Chicago Press, Chicago, 1962).

⁴T. Maudlin, *Philosophy of Physics: Quantum Theory* (Princeton University Press, Princeton, 2019); and *Philosophy of Physics: Space and Time* (Princeton University Press, Princeton, 2012).

theory's substantive physical claims from its mathematical formulation in ways that preserve the falsifiability picture at the level the picture is supposed to operate.⁵ The broader philosophy of physics community has continued to treat physics as the most rigorous empirical science available to human inquiry, with disagreements about specific cases sitting against a shared background commitment to the empirical-accountability framework.

This essay does not contest the standard view as a description of physics' aspirations or as an account of the discipline at its best. It accepts that physicists' substantive empirical claims—about specific particles, specific forces, specific cosmological parameters—are formulated in ways that admit of empirical test, and are revised when test results conflict with prior expectation. The General Relativistic prediction of light deflection at the 1919 eclipse, the prediction of the Higgs boson at the LHC, the discovery of the cosmic microwave background by Penzias and Wilson, the prediction and detection of gravitational waves by LIGO—these episodes are paradigmatic instances of the standard picture operating as advertised. Theory predicts; experiment tests; theory survives or is revised. The essay accepts that this is happening, in the cases where it happens.

What the essay raises is a structural asymmetry that operates beneath these surface successes. The substantive empirical claims of physics—the claims about particles, forces, fields, cosmological parameters, and the rest—are formulated within a mathematical apparatus that itself shapes what counts as a substantive empirical claim, what counts as a permissible theory-space within which such claims can be formulated, and what counts as anomaly versus what counts as feature. The mathematical apparatus is not itself a substantive empirical claim. It is the substrate within which substantive empirical claims become formulable. And this substrate is not subject to the falsifiability standard that physics applies to its substantive claims. The substrate is constrained by internal consistency requirements—by the demand that its deductions follow from its axioms, that its theorems not contradict one another, that its formal apparatus cohere with itself. It is not constrained by coherence requirements in the empirical sense—by the demand that its outputs match what experiment produces in the world.

⁵J. Earman, *Bangs, Crunches, Whimpers, and Shrieks: Singularities and Acausalities in Relativistic Spacetimes* (Oxford University Press, Oxford, 1995); and *World Enough and Space-Time: Absolute versus Relational Theories of Space and Time* (MIT Press, Cambridge, 1989).

This is the asymmetry the essay names and develops. The substantive claims of physics are subject to one epistemological standard; the mathematical substrate on which those claims rest is subject to a different and weaker standard. Both standards are reasonable in their own terms. Mathematical apparatus that was internally contradictory would be useless; mathematical apparatus whose external coherence-with-evidence requirement was elevated to the same status as its internal-consistency requirement would dissolve mathematics as a discipline, since mathematical objects are not themselves empirical and their truths are not vindicated by observation. The point of the essay is not that mathematics ought to be falsifiable in the empirical sense, which it cannot be. The point is that physics, in adopting a particular mathematical apparatus and treating that apparatus as the framework within which all physical questions must be posed, inherits both the consistency requirement and the absence of empirical-coherence requirement. The framework's choices about what kind of entities can exist, what kind of properties they can have, what kind of relations they can stand in, are not themselves subject to revision in the face of empirical anomaly. They are subject only to revision in the face of internal inconsistency, which the framework rarely exhibits because it has been designed to be internally consistent.

The consequence is that empirical anomalies in physics are routinely resolved by elaborations of the framework rather than by revisions of it. When the framework's outputs and the empirical record diverge, the empirical record is treated as a puzzle for the framework to eventually accommodate, rather than as evidence that the framework's assumptions about possible-entities and possible-relations may require revision. This is not a moral failing of physicists. It is a structural consequence of the framework's epistemic status: because the framework itself is not subject to empirical falsification, the natural response to disconfirming evidence is to seek a framework-internal accommodation rather than to question the framework. The accommodations accumulate. Each accommodation is locally reasonable. The cumulative effect is a framework increasingly elaborated, with increasingly elaborate auxiliary entities and increasingly elaborate auxiliary postulates, none of which can be falsified at the level the framework operates because the framework's mathematical structure forecloses on questions that would put the framework itself at risk.

The essay develops this asymmetry in five sections. Section II introduces the philosophical literature on the relationship between mathematics and physics, and locates the asymmetry within

that literature. Section III develops the central distinction the essay relies on: between consistency, the internal logical coherence of a formal system, and coherence, the correspondence between a framework's outputs and the empirical record. Section IV works through five cases in which the framework-preservation dynamic is visible: the historical case of the luminiferous aether (now uncontroversial in retrospect), the contemporary case of dark matter and dark energy, the cosmological constant problem, the proliferation of quantum gravity programs, and the symmetrization postulate in quantum mechanics. Section V draws the structural conclusion. Section VI argues that the philosophy of physics must take seriously the question of what alternative mathematical frameworks would look like, not as a matter of intellectual hobby but as a structural requirement for physics to remain accountable to evidence.

II. Mathematics and Physics: A Relationship Worth Examining

The relationship between mathematics and physics has been a topic of sustained philosophical attention since at least the early twentieth century, but the literature exhibits a striking pattern. Where philosophers have addressed the relationship, they have largely addressed two questions: why mathematics is so effective in describing physical reality, and what the ontological status of mathematical objects must be if such effectiveness is to be intelligible. The first question is associated with Wigner's famous essay on "the unreasonable effectiveness of mathematics in the natural sciences," in which he observed that mathematical concepts developed without reference to physical application have proven, with striking regularity, to be exactly the concepts physics turns out to need.⁶ The second question is associated with the Quine-Putnam indispensability argument, which holds that since mathematics is indispensable to our best scientific theories, and since we should be ontologically committed to the entities indispensable to our best theories, we should accept the existence of mathematical objects on broadly empiricist grounds.⁷ Maddy's subsequent naturalist response has refined this picture by distinguishing the practice of

⁶E. P. Wigner, "The Unreasonable Effectiveness of Mathematics in the Natural Sciences," *Communications in Pure and Applied Mathematics* 13, 1 (1960). The essay remains the standard reference for the question and continues to be cited extensively in the contemporary literature.

⁷W. V. O. Quine, "Two Dogmas of Empiricism," *Philosophical Review* 60, 20 (1951); H. Putnam, *Philosophy of Logic* (Harper Torchbooks, New York, 1971). The indispensability argument receives its canonical formulation across these works and their subsequent elaborations.

mathematics from the philosophical interpretation of that practice, arguing that mathematicians' own methodological commitments deserve primacy in any account of what mathematics is and how it functions in inquiry.⁸

These literatures are substantial and continue to develop. Pincock has recently offered a sustained analysis of how mathematical structures contribute to physical explanation, distinguishing the various roles mathematics plays in physical theories—representational, inferential, abstract—and developing an account of how each role contributes to explanatory success.⁹ Bueno and Colyvan have refined the indispensability argument and its critics, with attention to the specific ways mathematical apparatus enters into physical theorizing.¹⁰ French and Ladyman, working in the structural realist tradition, have argued that the structural features of mathematics carried by physical theories are themselves what the theories are committed to, with the question of what underlies the structure deferred or denied.¹¹ The relationship between mathematics and physics has been examined from multiple angles, and the examination has produced philosophical sophistication adequate to the topic as it is conventionally posed.

What this literature has examined less thoroughly is the question this essay raises. The standard treatments take for granted that mathematics, whatever its ontological status and however it achieves its effectiveness, is being deployed in physics as a transparent tool. The mathematics describes; physics tests the description against the world; the world either confirms or disconfirms; the description is preserved or revised accordingly. The mathematics is a medium of expression for physical claims; it is not itself a participant in the claims being expressed. On this picture, the mathematical apparatus is epistemologically neutral with respect to what physics finds. Whatever

⁸P. Maddy, *Naturalism in Mathematics* (Oxford University Press, Oxford, 1997); and *Second Philosophy: A Naturalistic Method* (Oxford University Press, Oxford, 2007). Maddy's mature view distinguishes the methodological autonomy of mathematics from the philosophical question of mathematical ontology.

⁹C. Pincock, *Mathematics and Scientific Representation* (Oxford University Press, Oxford, 2012).

¹⁰O. Bueno and M. Colyvan, "An Inferential Conception of the Application of Mathematics," *Noûs* 45, 345 (2011); M. Colyvan, *The Indispensability of Mathematics* (Oxford University Press, Oxford, 2001).

¹¹S. French and J. Ladyman, "Remodelling Structural Realism: Quantum Physics and the Metaphysics of Structure," *Synthese* 136, 31 (2003); J. Ladyman and D. Ross, *Every Thing Must Go: Metaphysics Naturalized* (Oxford University Press, Oxford, 2007).

physics discovers, the mathematics is positioned to represent it; whatever physics fails to discover, the mathematics is positioned to represent that absence as well.

This picture is not quite right, and the ways in which it is not quite right matter for the essay's central argument. Mathematical frameworks are not epistemologically neutral with respect to what they can be used to represent. They embody commitments about what kinds of entities exist, what kinds of properties such entities can have, what kinds of relations they can stand in. These commitments are not themselves derived from empirical inquiry; they are inherited from the framework's foundational choices, which were made for reasons largely internal to mathematics itself—for reasons of formal economy, of internal consistency, of generality, of fit with the surrounding mathematical apparatus that the framework needs to interoperate with. A framework built around discrete entities and their properties will permit certain kinds of physical theories and exclude others. A framework built around continuous fields and their values will permit different theories and exclude different others. A framework that treats spatial location as a primitive will permit different theories than one that treats interaction as primitive. The choice of mathematical framework is not posterior to physical inquiry; it is constitutive of what physical inquiry can find.

This observation is not novel in the philosophy of mathematics, where the constitutive role of mathematical frameworks has been examined in the foundations literature.¹² Nor is it novel in the philosophy of physics, where authors including Stein and Belot have noted that the mathematical structure of physical theories carries commitments not always foregrounded by the theories' physical interpretations.¹³ What has been less examined is the asymmetry this constitutive role introduces between the substantive empirical claims of physics and the mathematical framework within which those claims are formulated. The substantive claims are subject to falsification by empirical evidence. The framework within which they are formulated is not subject to falsification by empirical evidence, because the framework determines what counts as evidence in the first place. A framework that does not permit a certain kind of entity to exist will not register evidence

¹²P. Benacerraf, "What Numbers Could Not Be," *Philosophical Review* 74, 47 (1965); S. Shapiro, *Philosophy of Mathematics: Structure and Ontology* (Oxford University Press, Oxford, 1997).

¹³H. Stein, "Yes, but... Some Skeptical Remarks on Realism and Anti-Realism," *Dialectica* 43, 47 (1989); G. Belot, *Geometric Possibility* (Oxford University Press, Oxford, 2011); E. Knox, "Newtonian Spacetime Structure in Light of the Equivalence Principle," *British Journal for the Philosophy of Science* 65, 863 (2014).

for that entity as evidence; the evidence will be classified as anomaly, as measurement error, or as evidence for some entity the framework does permit. The framework selects what its outputs will be sensitive to, before any output is generated.

The implications of this asymmetry have been noted in passing by various authors but have not, to the essay's knowledge, been developed into a sustained philosophical analysis. Lakatos came closest, in his treatment of hard cores and protective belts; the hard core of a research programme, on his account, includes mathematical commitments that are protected from revision by the belt of auxiliary hypotheses around them.¹⁴ But Lakatos's framework was concerned with the empirical progressiveness of research programmes, not with the structural question of whether the hard core's mathematical commitments are themselves subject to empirical accountability in any meaningful sense. He acknowledged that they are not, and treated this as a description of how science actually works rather than as a problem requiring philosophical attention. The essay takes the same observation as Lakatos's starting point and treats it as a problem requiring attention.

The problem has the following structure. Physics, as a discipline, presents its claims as empirically accountable. Empirical accountability requires that the claims could in principle be falsified by evidence. Falsification by evidence requires that the framework within which the claims are formulated permits the evidence to register as evidence. If the framework does not permit a class of possible evidence to register, the claims formulated within the framework cannot be falsified by that class of evidence—not because the evidence does not exist, but because the framework forecloses on the question of whether it exists. The framework's choices about what kinds of entities and relations are possible are therefore prior to the empirical accountability physics claims for its substantive results. The framework is not itself empirically accountable. The substantive results that depend on it are accountable only within the framework. The accountability is real, but bounded; the boundary is set by the framework; the framework itself sits outside the accountability it makes possible for its outputs.

This is not yet an argument that the framework is wrong, or that physics is in any specific way deluded about what it is doing. It is an observation about the structural relationship between the framework and the substantive empirical claims. The observation creates a question: what happens

¹⁴Lakatos, *op. cit.*, particularly the discussion of the hard core's protection in sections 2 and 3 of the 1970 paper.

when the framework's structural commitments and the empirical record diverge? Two responses are available in principle. The first is to elaborate the framework—to add auxiliary entities, auxiliary postulates, auxiliary hypotheses that accommodate the divergence within the framework's existing commitments. The second is to revise the framework—to question whether the framework's structural commitments themselves require modification in light of the divergence. The essay's central claim, to be developed across the following sections, is that contemporary physics consistently selects the first response and consistently declines the second, and that this consistent selection is itself a structural feature of the discipline's epistemic situation, not a contingent feature subject to easy correction.

Section III develops the philosophical machinery the essay needs to make this argument precise. The machinery is the distinction between consistency and coherence, between the internal logical integrity of a formal system and the system's correspondence with the empirical record. Both are reasonable epistemic standards. They are not equivalent. When they diverge, a choice must be made about which to prefer. Contemporary physics, the essay will argue, has increasingly preferred consistency, and the preference has costs the field has not adequately acknowledged.

III. Consistency and Coherence

The distinction the essay relies on is between two epistemic standards that a body of claims can be expected to satisfy. The first standard, *consistency*, is internal. A body of claims is consistent if its members do not contradict one another, if its inferential apparatus produces no theorem that conflicts with another theorem, if its formal structure does not yield both a proposition and its negation as derivable. Consistency is a property a body of claims has with respect to itself. It is verified by examining the body of claims in isolation, checking that the inferential moves the body permits do not generate contradictions, and confirming that the axioms from which the body is generated do not themselves imply incompatible conclusions. A body of claims is consistent or it is not, and the question of its consistency is, in principle, settleable by formal examination alone.

The second standard, *coherence*, is external. A body of claims is coherent with the empirical record if its predictions match what experiment produces, if its descriptions accord with what observation discloses, if its outputs correspond to what the world delivers when interrogated. Coherence is a property a body of claims has with respect to something outside itself. It is verified by comparing

the body's outputs with measurements taken from a domain the body purports to describe, and confirming that the comparison yields agreement within whatever tolerances the inquiry has established as acceptable. A body of claims is coherent with the empirical record or it is not, and the question of its coherence is settled empirically, by inquiry into the domain the body claims to describe.

These two standards are distinct. A body of claims can be consistent without being coherent. Euclidean geometry is internally consistent—the proofs follow from the axioms, no theorem contradicts another—but is not coherent with the geometric structure of the physical universe at large scales, where general relativity tells us spacetime is curved and parallel lines do not preserve their parallelism. Euclidean geometry remains a perfectly good mathematical theory; what it is not is a description of how the physical universe geometrically behaves. A body of claims can also be coherent without being fully consistent, or at least without having its consistency fully verified. Early calculus achieved striking coherence with empirical mechanics for nearly two centuries before its foundations were rigorously developed; Newton's fluxions and Leibniz's infinitesimals were operationally effective long before they were given formally consistent treatment in the work of Cauchy and Weierstrass. The mathematics worked, in the sense of producing predictions that matched experiment, before anyone had demonstrated that the mathematics did not generate contradictions when used as Newton and Leibniz had used it.

Mathematics, as a discipline, prioritizes consistency. This is not a flaw in mathematics; it is constitutive of what mathematics is. Mathematical objects are not empirical objects; mathematical theorems are not empirical claims; mathematical proofs do not depend on observation for their warrant. The discipline's standards of validation are internal to its inferential structure, and rightly so. A mathematical claim is vindicated by being demonstrably derivable from accepted axioms by accepted inferential rules, not by being verified through experimental observation. Mathematics that fails to be consistent is mathematics that has failed by its own standards. Mathematics that fails to be empirically coherent is, on the discipline's own terms, not failing at all—mathematics was never offering empirical coherence as a deliverable.

Physics, as a discipline, prioritizes coherence—or at any rate claims to. Physical theories are vindicated by experimental confirmation; they are revised when experiment delivers results the theory did not predict; their warrant depends on their continuing to match what the world produces

under controlled interrogation. A physical theory that is internally consistent but empirically incoherent has failed; the inconsistency between its predictions and the experimental record is reason to revise it, regardless of how elegant or formally satisfying its mathematical structure might be. This commitment to coherence over consistency, when the two diverge, is what distinguishes physics from mathematics as a discipline. Mathematics asks: does the argument follow? Physics asks: does the world behave as the argument predicts? The two questions have different criteria of success, and the disciplines have been organized accordingly.

The trouble begins when physics adopts a mathematical apparatus as its medium of expression. The apparatus is, of necessity, mathematical—it inherits the consistency standard of mathematics, since a mathematical apparatus that was internally inconsistent would be useless even as a representational tool. But the apparatus also carries commitments: about what kinds of entities exist, what kinds of properties they can have, what kinds of relations they can stand in. These commitments are inherited from the mathematics' foundational choices, which were made for reasons internal to mathematics. The commitments are protected by the consistency standard—they cannot be revised in ways that introduce contradiction. They are not, however, protected by the coherence standard, because mathematics does not operate under a coherence standard. When the apparatus's commitments and the empirical record diverge, the apparatus itself is not at risk; what is at risk is the substantive theory built on the apparatus.

The asymmetry this creates is the structural fact the essay identifies. The substantive theory is subject to coherence: it must match the empirical record, or be revised. The apparatus is subject only to consistency: it must be internally non-contradictory, but it need not match anything outside itself. When the substantive theory and the empirical record diverge, the response is to revise the theory, not the apparatus. When such revisions accumulate—when the substantive theory has been repeatedly elaborated to accommodate empirical surprises while the apparatus remains untouched—the discipline has, in effect, treated the apparatus as more secure than the empirical record itself. The apparatus survives unchanged across surprises that, if directed at the substantive theory alone, would have prompted revision. The apparatus has been protected from the coherence standard by being treated as part of mathematics rather than as part of physics.

This is the central diagnostic the essay offers. Physics presents itself as a coherence-prioritizing discipline. In practice, the discipline operates with two layers: a substantive theory layer at which

the coherence standard applies, and a mathematical apparatus layer at which only the consistency standard applies. The two layers are entangled—the substantive theory is formulated within the apparatus, and the apparatus shapes what counts as a permissible substantive theory—but they are subject to different epistemic standards. The substantive theory must be coherent with evidence; the apparatus need only be consistent with itself. When divergence between framework outputs and evidence arises, the discipline’s structural response is to seek revisions at the substantive layer that preserve the apparatus, because revisions at the apparatus layer would require giving up commitments that the discipline does not have a procedure for revising. The apparatus is, in effect, treated as background dogma—not because anyone has decided to treat it as dogma, but because the discipline’s two-layer structure makes the apparatus immune to the standard that would otherwise hold it accountable.

The word *dogma* is strong, and the essay uses it with care. It does not mean that the apparatus is arbitrary, that its choices were made without reason, or that mathematicians lacked good motives for the choices they made. The apparatus is, in its own terms, a remarkable intellectual achievement—internally consistent, generative, capable of representing extraordinary varieties of structure, refined across centuries of mathematical work. Calling it dogma is not a judgment about the apparatus’s mathematical merits. It is a judgment about its epistemic status within physics: it is treated as a fixed starting point rather than as a hypothesis subject to revision in light of evidence. The apparatus’s mathematical merits are what they are; the question is whether the discipline that adopted the apparatus as its medium has retained the capacity to question the adoption when the adoption produces sustained divergence from evidence.

The standard response to this diagnostic is to deny the asymmetry. The mathematical apparatus of physics, the response holds, is not dogma; it is whatever apparatus has proven most empirically successful. If the apparatus stopped being empirically successful, the discipline would adopt a different apparatus. The current apparatus is therefore not protected from empirical revision; it has been empirically vindicated, repeatedly, across the long history of physics’ successes.

This response is half right. The mathematical apparatus of physics has been empirically vindicated at the level of the substantive theories built on it; the predictions of general relativity, the predictions of quantum field theory, the calculations that produced the standard model and its various confirmations have been spectacular successes by any reasonable standard. What the

response elides is the structural question: would the discipline recognize the apparatus's failure if the apparatus were failing? The discipline's structural response to divergence is to revise the substantive theory while preserving the apparatus. If the apparatus is failing, the response that would detect the failure is precisely the response the discipline does not perform. Empirical vindication, on this picture, is a vindication of the substantive theory built on the apparatus, and at most an indirect vindication of the apparatus itself. The apparatus has been empirically vindicated in the sense that some substantive theory formulated within it has succeeded. Whether a substantive theory formulated within a different apparatus would succeed at the questions the current apparatus has struggled with is a question the current discipline does not have the procedure to answer, because answering it would require taking the apparatus itself as a hypothesis rather than as a starting point.

The essay's substantive cases, developed in the next section, are intended to demonstrate that the structural pattern just described is operating in identifiable contemporary physics. Each case is one in which the apparatus's commitments and the empirical record have diverged; in which the discipline's response has been to elaborate substantive theory rather than to revise the apparatus; and in which the elaborations have accumulated to a degree that, on a coherence-prioritizing standard, would justify questioning the apparatus itself. The cases are not chosen as gotchas. They are chosen because each one has been extensively documented in the physics literature, each one is recognized as a substantive puzzle within the field, and each one exhibits the framework-preservation dynamic at a scale that makes the dynamic visible.

IV. The Pattern in Practice

The argument of the preceding section is structural: physics operates as a two-layer system in which the substantive theory layer is subject to coherence with evidence and the mathematical apparatus layer is subject only to internal consistency. When the two layers diverge, the discipline's structural response is to revise the substantive theory while preserving the apparatus. The argument has been developed at the conceptual level. The cases that follow demonstrate the structure operating in identifiable contemporary physics. Each case exhibits the same pattern: empirical divergence from framework predictions, followed by accommodation through substantive-theory elaboration rather than through framework revision, followed by accumulation

of elaborations to a degree that, on a coherence-prioritizing standard, would justify questioning the framework itself.

The cases are presented in order of increasing contemporaneity, beginning with a historical case whose resolution is now uncontroversial and proceeding to contemporary cases whose status remains contested. The order is deliberate. The historical case establishes the pattern's shape in a setting where hindsight has confirmed which response was correct; the contemporary cases then become recognizable instances of the same shape playing out before the resolution has occurred.

A. The Luminiferous Aether

The luminiferous aether was, for most of the nineteenth century, the medium through which light was understood to propagate. The mathematical apparatus of nineteenth-century optics, modeled on the mechanics of waves in elastic media, required a substrate. Sound waves propagate through air; water waves propagate through water; if light is a wave, the reasoning ran, light must propagate through something. The aether was the entity the mathematical apparatus required, and its properties were specified by what the apparatus needed it to be: rigid enough to support the high frequencies of visible light, yet diffuse enough to permit the unimpeded motion of planets through it. The properties were peculiar but unavoidable, given the framework's commitments about how wave propagation must work.¹⁵

When Michelson and Morley failed to detect Earth's motion through the aether in their famous interferometer experiments of the 1880s, the standard response was not to question the framework's commitment to a propagating medium. The standard response was to elaborate the framework: to postulate that the aether dragged along with Earth, then that the dragging was partial, then that bodies moving through the aether were physically contracted in the direction of motion in just the way required to mask the aether's effect. Lorentz developed the contraction hypothesis as a substantial mathematical apparatus, deriving the transformation equations that bear his name and showing that the apparatus could accommodate Michelson and Morley's null result

¹⁵A. Whittaker, *A History of the Theories of Aether and Electricity*, 2nd edition (Thomas Nelson, London, 1951). The work remains the standard historical reference for the aether period and its eventual displacement.

while preserving the aether and the framework that required it.¹⁶ FitzGerald independently arrived at a similar proposal. The aether was preserved; the framework that required it was preserved; the empirical anomaly was accommodated through elaboration of the substantive theory built on the framework.

Einstein's 1905 paper on the electrodynamics of moving bodies did not introduce new empirical evidence against the aether. The Michelson-Morley result was already known. What Einstein introduced was a framework revision: the principle of relativity and the constancy of the speed of light, taken as primitive postulates, with the Lorentz transformations rederived as consequences rather than as ad hoc accommodations.¹⁷ On Einstein's framework, there was no aether to detect because the framework no longer required one. The empirical predictions of Einstein's framework matched Lorentz's framework in the cases Lorentz had constructed his accommodations to handle, and exceeded Lorentz's framework in cases the accommodations had not anticipated.

The lesson of this case, for the structural argument of this essay, lies not in Einstein's specific framework but in the structural situation that preceded its adoption. For roughly two decades after Michelson and Morley, the empirical record was already in conflict with the framework's commitments. The discipline's response was to elaborate. The elaboration was ingenious, mathematically rigorous, and internally consistent. It was also, in retrospect, a sustained exercise in framework-preservation when framework-revision was the response coherence with evidence required. The discipline reached the framework-revision response only when an outsider—a patent clerk without institutional position in physics—performed the revision in a paper that did not initially attract the attention its content deserved. The pattern is recognizable: extended elaboration in defense of the framework, eventual revision arriving from a direction the discipline had not anticipated.

¹⁶H. A. Lorentz, "Michelson's Interference Experiment," *Versuch einer Theorie der elektrischen und optischen Erscheinungen in bewegten Körpern* (Brill, Leiden, 1895). The contraction hypothesis is developed in §89–§92.

¹⁷A. Einstein, "Zur Elektrodynamik bewegter Körper," *Annalen der Physik* 17, 891 (1905). English translation widely available.

The case is uncontroversial today because the resolution is known. What this essay asks the reader to consider is whether the same pattern, recognizable in retrospect for the aether, may be operating in contemporary physics without yet having been resolved.

B. Dark Matter and Dark Energy

The gravitational dynamics of galaxies, galaxy clusters, and the large-scale structure of the cosmos do not match what general relativity predicts on the basis of visible matter alone. Galactic rotation curves remain flat at large radii rather than declining as Keplerian dynamics would require. Galaxy clusters exhibit gravitational binding stronger than their visible mass should produce. Observations of the cosmic microwave background and of large-scale structure formation require approximately five times more matter than is visible, plus a separate component of “dark energy” driving accelerating cosmic expansion. The discrepancies between framework predictions and empirical observation are not subtle; they are factor-of-five disagreements about the mass content of the universe and factor-of-many-orders-of-magnitude disagreements about the energy density of the vacuum.¹⁸

This case differs from the aether case in an important respect that deserves explicit acknowledgment. The evidence for a gravitational anomaly is not confined to a single experimental result; it converges from multiple independent lines of investigation. Galaxy rotation curves, gravitational lensing including the colliding-cluster cases such as the Bullet Cluster, the acoustic peak structure of the cosmic microwave background, and the statistics of large-scale structure formation all require, within the standard framework, an additional gravitating component. The aether had essentially zero independent evidence beyond its role in the mathematical apparatus of wave propagation; the gravitational anomaly does not have this property. Multiple convergent probes point to the existence of an empirical phenomenon that

¹⁸M. S. Turner and J. A. Tyson, "Cosmology at the Millennium," *Reviews of Modern Physics* 71, S145 (1999); P. J. E. Peebles and B. Ratra, "The Cosmological Constant and Dark Energy," *Reviews of Modern Physics* 75, 559 (2003); for an accessible synthesis, R. H. Sanders, *The Dark Matter Problem: A Historical Perspective* (Cambridge University Press, Cambridge, 2010). The convergence of independent probes — galaxy rotation curves, gravitational lensing including the Bullet Cluster, the CMB acoustic peaks, and large-scale structure formation — is reviewed across these references.

requires explanation. This is the strongest empirical situation the essay considers, and it is the case in which the structural argument must be made most carefully.

The careful version of the argument distinguishes three levels of claim. At the phenomenon level, an additional gravitating component is required by the empirical record with high confidence. At the model level, the standard framework accommodates this requirement by parameterizing the additional component as a non-baryonic, non-electromagnetically-interacting form of matter together with a cosmological-constant or dark-energy term, producing the standard Λ CDM model that fits the observational data with high precision. At the ontic level, the claim that the additional gravitating component is constituted by a specific kind of particulate entity—that it consists of particles with definite mass, interaction cross-section, and abundance—is a further commitment that the empirical record does not, by itself, establish. The phenomenon is robust; the parameterized model is empirically successful within the standard framework; the ontic interpretation of the model's additional component as a particle species is the layer at which the framework-dependence becomes visible.

The discipline's response to the phenomenon has been to postulate dark matter as an entity at the ontic level. Dark matter is, by stipulation, matter that does not interact electromagnetically—it neither emits nor absorbs light, neither reflects nor scatters it, neither couples to the photon field nor exhibits any of the interactions through which ordinary matter has historically been detected. It interacts gravitationally, because the gravitational effects are what we are trying to explain. Beyond gravitational interaction, its properties are largely specified by what the framework requires: it must be cold (slow-moving relative to light at the relevant epoch), it must not interact strongly enough to dissipate energy and form structures the way ordinary matter does, and it must exist in approximately a five-to-one ratio with ordinary matter.¹⁹ Dark energy is even more sparingly characterized: it has whatever equation of state is needed to produce the observed cosmic acceleration, with the simplest version being a cosmological constant whose value is specified by observation rather than derived from theory.

¹⁹G. Bertone, D. Hooper, and J. Silk, "Particle Dark Matter: Evidence, Candidates and Constraints," *Physics Reports* 405, 279 (2005). The review remains a standard reference; subsequent developments have not displaced its central characterization of the candidate landscape.

Direct detection efforts for dark matter have been underway for over four decades. Increasingly sensitive detectors—including XENON, LUX, PandaX, and most recently LZ—have searched for the weakly interacting massive particles (WIMPs) that the leading dark matter candidates predict. The search has covered the parameter space the leading candidates occupy with increasing thoroughness. No direct detection has been confirmed. The search has not falsified the dark matter hypothesis, because the hypothesis can be elaborated to place the particles in regions of parameter space not yet probed. Each successive non-detection has been accommodated by revising the parameter ranges within which the particles are expected to lie; the framework's commitment to a particulate ontology has not been revised by the accumulated non-detections.²⁰

Modified gravity approaches, which propose framework revision at the level of the gravitational dynamics rather than addition of a new entity, have been pursued by a small but persistent minority of researchers. Milgrom's MODified Newtonian Dynamics (MOND) and its relativistic extensions account for galactic rotation curves with parametric economy that the dark-matter approach does not match, though MOND faces its own difficulties at galaxy-cluster scales and with cosmological structure formation.²¹ The MOND program has been treated as a fringe pursuit by the mainstream of the field, despite its empirical successes at the scales where dark matter does its primary explanatory work. The treatment reflects the structural fact identified in section III: the framework's commitment to general relativity in its current mathematical form is protected from revision; the response to anomaly is to elaborate substantive theory (postulating new entities) rather than to revise the framework (modifying the gravitational dynamics themselves), and such modified-gravity programs as exist are treated as fringe pursuits despite their parametric economy.

The analogy to the aether case is therefore structural rather than literal. The claim is not that dark matter does not exist as a phenomenon in the way the aether did not exist as a substance; the

²⁰For the experimental status, see the periodic reviews published in the *Annual Review of Nuclear and Particle Science*; for a recent comprehensive assessment of WIMP non-detection, J. Aalbers et al. (LZ Collaboration), "First Dark Matter Search Results from the LUX-ZEPLIN (LZ) Experiment," *Physical Review Letters* 131, 041002 (2023).

²¹M. Milgrom, "A Modification of the Newtonian Dynamics as a Possible Alternative to the Hidden Mass Hypothesis," *Astrophysical Journal* 270, 365 (1983); J. D. Bekenstein, "Relativistic Gravitation Theory for the Modified Newtonian Dynamics Paradigm," *Physical Review D* 70, 083509 (2004); for the program's contemporary status, B. Famaey and S. S. McGaugh, "Modified Newtonian Dynamics (MOND): Observational Phenomenology and Relativistic Extensions," *Living Reviews in Relativity* 15, 10 (2012).

gravitational anomaly is empirically robust in a way the aether-as-medium-of-light-propagation never was. The claim is rather that the framework's commitment to interpreting the gravitational anomaly through a particular ontology—particulate non-baryonic matter—performs structural work that the empirical record by itself does not require. The phenomenon establishes that something additional is gravitationally present; the framework determines that this something must be parsed as a particle species, because the framework's commitments admit this interpretation while excluding alternative interpretations that a different mathematical apparatus might license. The structural-preservation pattern operates at the ontic interpretation of the phenomenon, not at the existence of the phenomenon itself. This is the precise location of the framework-preservation dynamic in the dark matter case: the discipline preserves not merely the substantive theory but the ontological framework within which the empirical anomaly must be interpreted, and the four decades of failed direct detection are framework-internal accommodations of an ontological commitment whose grounding lies in the framework rather than in the empirical record.

C. The Cosmological Constant Problem

The vacuum energy density predicted by quantum field theory, on the basis of zero-point energies of the quantum fields that the framework requires, exceeds the observed cosmological constant by approximately 120 orders of magnitude. The discrepancy is, by a substantial margin, the largest known disagreement between theoretical prediction and empirical measurement in any physical science. The standard characterization in the physics literature is that this is “the worst prediction in theoretical physics.”²²

The structural significance of how this characterization functions is worth examining. A 120-order-of-magnitude disagreement between framework prediction and empirical observation, if it occurred in any substantive physical theory not protected by framework status, would be treated as definitive falsification. No theory whose predictions diverged from observation by 120 orders of magnitude would survive serious scientific consideration. The cosmological constant discrepancy survives consideration not because its magnitude has been disputed—it has not—but because the framework that generates the disagreeable prediction is treated as a starting point that

²²S. Weinberg, "The Cosmological Constant Problem," *Reviews of Modern Physics* 61, 1 (1989). The characterization of the problem as the worst prediction in theoretical physics is widely attributed to Weinberg's review and has become standard.

must be preserved, with the discrepancy reclassified as a “problem” the framework will eventually solve rather than as evidence the framework’s assumptions about vacuum structure require revision.

The proposed solutions to the cosmological constant problem are instructive. Anthropic reasoning holds that we observe a small cosmological constant because only in universes with small cosmological constants could observers exist to do the observing; the framework’s prediction is rescued by appeal to a multiverse in which the prediction is correct for most regions but observers are confined to a region where the prediction happens to fail.²³ Fine-tuning proposals seek mechanisms by which the framework’s prediction is mostly correct but a precise cancellation reduces the net result by 120 orders of magnitude. Supersymmetric proposals invoke partner particles whose contributions cancel against the predicted contributions in just the right way. Each proposed solution preserves the framework’s commitment to quantum field theory as currently formulated and locates the failure of prediction in some additional structure that has not yet been identified.

What none of the leading proposals do is reconsider whether the framework’s calculation of vacuum energy might rest on assumptions that the empirical record is telling us are wrong. The calculation depends on treating the vacuum as the ground state of the quantum fields, with all modes of all fields contributing their zero-point energies, summed across the relevant momentum range. Each step in this calculation is a framework commitment: that the relevant entities are quantum fields, that they have ground states, that ground-state energies sum in the manner the calculation assumes, that the momentum range is what the calculation assumes. The 120-order-of-magnitude discrepancy with observation is, on a coherence-prioritizing reading, evidence that one or more of these framework commitments may require revision. The discipline does not read it this way. It reads the discrepancy as a problem within the framework, requiring solution from within the framework’s resources.

D. The Proliferation of Quantum Gravity Programs

²³S. Weinberg, “Anthropic Bound on the Cosmological Constant,” *Physical Review Letters* 59, 2607 (1987); for the contemporary anthropic landscape, L. Susskind, *The Cosmic Landscape: String Theory and the Illusion of Intelligent Design* (Little, Brown, New York, 2006).

For approximately seventy-five years, physics has lacked a consistent theoretical framework that unifies general relativity (the theory of gravity at large scales) with quantum field theory (the theory of matter at small scales). The two frameworks employ incompatible mathematical structures: general relativity treats spacetime as a dynamical entity whose geometry is determined by the matter content of the universe, while quantum field theory treats spacetime as a fixed background on which the matter content lives. Combining them in any naive way produces mathematical inconsistencies; producing a consistent combination has been the goal of the quantum gravity research program since the late 1940s.²⁴

The result of seventy-five years of effort has been the proliferation of competing programs. String theory, in its various formulations, treats elementary entities as one-dimensional extended objects in higher-dimensional spaces. Loop quantum gravity treats spacetime itself as quantized at the Planck scale, with discrete structure beneath the apparent continuum. Causal set theory treats spacetime as a discrete partial order of events. Asymptotic safety holds that gravity becomes a well-defined quantum field theory at sufficiently high energies. Emergent gravity programs propose that gravity is not fundamental but arises from underlying microphysical structure. Each program has produced extensive mathematical apparatus, internally consistent, and each has produced predictions that are too distant from current experimental access to permit empirical adjudication between them.²⁵ To be sure, some programs offer more concrete phenomenological signatures than others, but none has reached the level of empirical adjudication that would force a choice among them.

The structural significance of this situation has been remarked upon by Smolin and others sympathetic to the field, who have noted that a research program lasting seventy-five years without empirical adjudication has, by ordinary scientific standards, exhibited something other than the

²⁴For an accessible historical overview, C. Rovelli, "Notes for a Brief History of Quantum Gravity," in *Recent Developments in Theoretical and Experimental General Relativity, Gravitation, and Relativistic Field Theories*, edited by V. G. Gurzadyan, R. T. Jantzen, and R. Ruffini (World Scientific, Singapore, 2002), pp. 742–768.

²⁵For the contemporary state of the competing programs, J. Polchinski, *String Theory*, Vols. 1 and 2 (Cambridge University Press, Cambridge, 1998); C. Rovelli, *Quantum Gravity* (Cambridge University Press, Cambridge, 2004); R. D. Sorkin, "Causal Sets: Discrete Gravity," in *Lectures on Quantum Gravity*, edited by A. Gomberoff and D. Marolf (Springer, New York, 2005), pp. 305–327; M. Niedermaier and M. Reuter, "The Asymptotic Safety Scenario in Quantum Gravity," *Living Reviews in Relativity* 9, 5 (2006); E. P. Verlinde, "On the Origin of Gravity and the Laws of Newton," *Journal of High Energy Physics* 04, 029 (2011).

empirical progressiveness Lakatos would require of a healthy research programme.²⁶ The standard response to this observation has been that quantum gravity addresses phenomena at energies sufficiently extreme that current experiments cannot reach them, and that empirical adjudication must therefore await technologies not currently available. The response is not unreasonable. What it elides is the structural question: the proliferation of mathematically consistent but empirically indistinguishable frameworks is itself a feature of the situation worth examining. Each program preserves a different subset of the existing framework's commitments while modifying others. The proliferation indicates that the framework's commitments admit many different formal continuations, none uniquely required by the existing empirical record, and that the choice among continuations is being made on grounds other than empirical adjudication—on grounds of mathematical elegance, of programmatic ambition, of researcher communities' inherited training.

This is not, in itself, evidence that the existing framework is wrong. It is, however, an instance of the asymmetry the essay has identified. A consistency-prioritizing discipline can sustain multiple internally consistent formal continuations without resolution; a coherence-prioritizing discipline could not, because the coherence standard would force adjudication when the empirical record diverged from any given program's predictions. The fact that quantum gravity has sustained five or more major competing programs across seven decades without empirical adjudication is a sign of the consistency-prioritizing structure operating in full visibility.

E. The Symmetrization Postulate

A fifth case, briefer than the others, illustrates the pattern in a setting more elementary than cosmology or quantum gravity. The framework of standard quantum mechanics treats particles as labeled entities that can in principle be distinguished from one another, even when they share all of their measurable properties. The empirical record does not cooperate. Identical particles—two electrons in an atom, two photons in a cavity, two helium-4 nuclei in a condensate—cannot be distinguished by any experimental procedure. The framework's commitment to labeled entities and the empirical record's refusal to honor the labels are in immediate conflict.

²⁶L. Smolin, *The Trouble with Physics: The Rise of String Theory, the Fall of a Science, and What Comes Next* (Houghton Mifflin, Boston, 2006). Smolin's diagnostic is more focused on string theory specifically, but the structural point about empirical adjudication applies more broadly.

The framework's response is the symmetrization postulate. The postulate stipulates that, for systems of identical particles, the wave function must be either symmetric or antisymmetric under exchange of any two particles' labels. The postulate is added to the framework's standard machinery as an additional postulate, beyond the foundational axioms of quantum mechanics, and is justified by its empirical success: with the postulate, quantum mechanics produces predictions that match what experiment finds with identical particles.²⁷

The structural reading of this situation is that the framework's commitment to labeled entities has been preserved at the cost of an ad hoc postulate whose function is to neutralize the empirical content of the labels. The postulate does not eliminate the labels from the formalism; the labels remain, and operations on the wave function are still performed with reference to them. What the postulate does is require that all physical predictions be symmetric under permutation of the labels, with the consequence that the labels can never appear in any physical prediction. The empirical content of the framework—what it predicts experiment will find—is precisely the content a framework without labels would predict. The framework, however, retains the labels because removing them would require revising the framework at its foundations rather than at its postulate-list.

The symmetrization postulate is, on a coherence-prioritizing reading, evidence that the framework's commitment to labeled entities is the wrong commitment to make. A framework that started from indistinguishability rather than distinguishability—that treated the absence of distinguishing features as primary rather than derivative—would not require the postulate. The framework would predict identical-particle behavior as a feature, not need to legislate it as a postulate.²⁸ The standard framework does not take this route, because doing so would require framework revision rather than postulate addition.

²⁷For the symmetrization postulate's standard treatment, J. J. Sakurai and J. Napolitano, *Modern Quantum Mechanics*, 2nd edition (Addison-Wesley, San Francisco, 2011), §7.1. For sustained analysis of how framework commitments shape interpretation across quantum mechanics, D. Wallace, *The Emergent Multiverse: Quantum Theory according to the Everett Interpretation* (Oxford University Press, Oxford, 2012).

²⁸For an alternative-framework approach to particle indistinguishability that develops this point in detail, E. Arete, *Particle Indistinguishability as the Empirical Warrant for a Relational Ontology* (Zenodo, 2026), <https://doi.org/10.5281/zenodo.20486256>. The structural reading of the fermion/boson distinction is further developed in E. Arete, *Inscription, Admittance, and Coherence: A Structural Reading of the Fermion/Boson*

F. The Pattern

The five cases differ in their specifics. The aether is a historical case whose resolution is now uncontroversial. Dark matter and dark energy are contemporary cases at the limits of empirical investigation, with the framework-preservation dynamic visible at the ontic-interpretation layer of an empirically robust phenomenon. The cosmological constant problem is the case where the framework's quantitative failure is most extreme. The quantum gravity proliferation is the case where the absence of empirical adjudication is most prolonged. The symmetrization postulate is the case where the framework's structural commitment is most directly in tension with the empirical record on elementary grounds.

What the cases share is the structural pattern the essay has named. In each case, the framework's commitments and the empirical record have diverged. In each case, the discipline's response has been to elaborate substantive theory while preserving the framework. In each case, the elaboration has accumulated to a degree that, on a coherence-prioritizing standard, would justify questioning the framework itself. In each case, the questioning has not occurred at the level the structural situation suggests it should—or, as in the aether case, has occurred only after an extended delay and through intervention from a direction the discipline did not anticipate.

The next section draws the structural conclusion these cases make available. The conclusion is not that the framework is definitively wrong, that dark matter does not exist as a phenomenon, that the symmetrization postulate is unwarranted, or that any specific elaboration is illegitimate. The conclusion is structural: the pattern these cases share is a pattern, and the pattern is what the asymmetry between consistency and coherence makes possible. The pattern can continue indefinitely without internal pressure for revision. Whether physics, as a discipline, will eventually achieve the framework-revision response that the structural situation suggests it should is a question the discipline's two-layer structure does not provide the resources to answer.

V. The Structural Conclusion

Distinction at the Elementary Level (Zenodo, 2026), <https://doi.org/10.5281/zenodo.20486258>. For background, see also S. French and D. Krause, *Identity in Physics: A Historical, Philosophical, and Formal Analysis* (Oxford University Press, Oxford, 2006).

The cases of section IV have been presented in their specifics, but their cumulative significance lies in the structural pattern they collectively exhibit. The pattern is not subtle once it has been named, and it is worth stating in its strongest defensible form.

Across each of the cases examined, the framework's commitments and the empirical record have diverged. In each case, the discipline's response to the divergence has been to elaborate the substantive theory built on the framework, while preserving the framework's foundational commitments. In the aether case, the elaboration took the form of dragging coefficients and length contractions; in the dark matter case, the elaboration takes the form of an ontic commitment to a particulate interpretation of a gravitational anomaly, sustained across four decades of failed direct detection; in the cosmological constant case, the elaboration takes the form of anthropic reasoning, multiverse postulation, and fine-tuning mechanisms; in the quantum gravity case, the elaboration takes the form of multiple competing internally consistent formulations whose empirical distinguishability lies beyond current experimental access; in the symmetrization postulate case, the elaboration takes the form of an ad hoc postulate whose function is to neutralize the empirical content of a framework commitment the empirical record does not honor. The specifics differ. The structural pattern is identical.

What the pattern indicates is not that physics is producing bad theories or that physicists are reasoning poorly. The substantive theories built on the framework remain, in many cases, extraordinarily successful at the levels the framework operates well. General relativity continues to predict gravitational waves and gravitational lensing and time dilation with extraordinary precision in the regimes where the framework's commitments and the empirical record do not diverge. Quantum field theory continues to produce predictions in particle physics that match experimental observation to extraordinary precision in the regimes where the framework operates as designed. The framework has been astonishingly productive.

The asymmetry the essay names is, accordingly, more refined than a flat insulation of the framework from coherence with evidence. The framework is subject to coherence in the regimes where its substantive predictions are tested and confirmed; in such regimes, the framework's commitments receive empirical vindication along with the theories built on them. The asymmetry is rather that successes accrue to the framework as a whole while failures are absorbed by the substantive theory layer alone. This asymmetry is not unreasonable as applied to any single

empirical divergence—a single failed prediction underdetermines whether the framework or the substantive theory built on it requires revision, and routing the revision to the substantive theory layer is the structurally conservative response when the source of the failure is unclear. The asymmetry becomes structurally consequential, however, when divergences accumulate across multiple independent cases, as the essay has documented across the five cases of section IV. The same two-layer system that produces both the successes and the failures is asked to take credit for one and assign blame for the other. This is a credit-assignment asymmetry rather than a pure insulation from test. The framework is shielded not by being beyond empirical evaluation but by being structurally positioned to receive credit while declining blame. Where the framework's commitments and the empirical record agree, the framework is treated as having predicted the agreement; where the commitments and the record disagree, the framework is treated as having provided the medium within which the discrepancy can eventually be resolved through elaborations of the substantive theory layer. Vindication is at the framework level; failure is at the substantive level. The asymmetry is the credit-assignment rule, and the rule is what makes sustained empirical pressure on the framework difficult to apply across accumulated cases even when it would be reasonable to apply on any single one.

The exemption has costs. The first cost is the accumulation of elaborations. Each individual elaboration is locally reasonable: postulating dragging coefficients to accommodate Michelson and Morley was a reasonable response, given the framework's prior commitments; postulating a particulate ontology to accommodate galactic rotation curves is a reasonable response, given the framework's prior commitments; postulating anthropic selection to accommodate the cosmological constant is a reasonable response, given the framework's prior commitments. The reasonableness of each elaboration, taken individually, is what makes the cumulative pattern hard to recognize from within the discipline. The pattern is visible only when the elaborations are taken together, and even then it is visible only against the backdrop of the structural argument the essay has been developing. From inside the discipline, the elaborations are the work of physics. From outside, the elaborations form a series whose direction is identifiable.

The second cost is the foreclosure of alternatives. A framework whose elaborations are protected from coherence testing cannot be displaced by a framework with superior coherence properties, because the displacement would require comparing the two frameworks against the empirical

record, and the comparison is the comparison the protected framework's structural status forecloses. Alternative frameworks can be proposed; they have been proposed, in various forms, throughout the history of physics. What they cannot do, against a framework protected from coherence testing, is win a coherence-based contest. The contest cannot occur on those grounds because the protected framework's grounds for retention are not coherence-grounds. The retention is grounded in consistency, in productivity in non-divergent regimes, in mathematical elegance, and in institutional continuity. Alternative frameworks must therefore win on the protected framework's grounds, which they cannot, because the protected framework was constructed to win on those grounds and the alternatives, being alternatives, were not.

The third cost is the gradual disconnection of the framework from the empirical accountability the discipline claims for itself. A framework that has accumulated decades of unfalsifiable elaborations is not, in any operational sense, accountable to evidence at the framework's own level. It is accountable to evidence at the level of its substantive predictions in non-divergent regimes; it is not accountable to evidence at the level of its foundational commitments. The discipline's self-presentation as empirically accountable continues to be supported by the substantive-predictions level. The structural reality at the foundational level differs from the self-presentation. The discipline operates, increasingly, as two enterprises: an empirically accountable enterprise at the level of substantive prediction within established regimes, and a consistency-prioritizing enterprise at the level of foundational commitment. The two enterprises share personnel, share institutional infrastructure, share a self-understanding as a single discipline. Structurally, they are not a single discipline. They are two enterprises operating under one name.

What the essay's argument does not claim is that the structural situation can be resolved by an act of will. Physicists are not deluded about what they are doing; they are operating in a structural situation whose features the essay has tried to describe. The structural situation does not provide an obvious procedure for its own correction, because the procedure that would correct it—questioning the framework when its empirical divergences accumulate—is precisely the procedure the structural situation does not have the resources to perform on itself. The framework's exemption from coherence testing is not a policy that can be repealed by a journal editor or a funding agency or a famous physicist. It is a structural fact about how the two-layer system

operates. Resolving it would require a change in the operating structure, not a change in the dispositions of the people operating within the structure.

What the essay's argument does claim is that the structural situation deserves more philosophical attention than it has received. The philosophy of physics has examined many aspects of the framework's commitments—what kinds of entities the framework permits, what kinds of properties they can have, what kinds of relations they can stand in, what the framework's interpretations imply about the nature of physical reality. What the philosophy of physics has examined less thoroughly is the structural question of why the framework's commitments are not themselves subject to the empirical accountability the discipline claims for its substantive predictions. The question is not a question internal to physics; physics does not have a procedure for asking it. The question is a question for the philosophy of physics, because the philosophy of physics is the discipline whose subject matter is what physics does and how it does it. The asymmetry between substantive predictions and foundational commitments is part of what physics does, and how it does it, and the philosophy of physics is positioned to examine the asymmetry in a way physics is not.

The next section addresses what the examination might productively look like. The argument is not that alternative frameworks must be adopted, that any specific alternative is superior, or that the existing framework must be discarded. The argument is more modest: that the philosophy of physics must take seriously the question of what alternative mathematical frameworks would look like, because doing so is the only structural intervention that could correct the asymmetry the essay has identified. The intervention is not an intervention against physics; it is an intervention in the conditions under which physics can be empirically accountable at its foundations rather than only at its surface.

VI. The Necessity of Alternative-Framework Inquiry

The structural conclusion of the previous section establishes the diagnostic position of the essay. The present author's recognition of this asymmetry arose during independent work developing an alternative axiomatic framework, cited later in this section; the asymmetry itself, however, stands independent of any specific alternative. The discipline of physics, in its current operating structure, has limited capacity to perform framework revision in response to the accumulation of empirical

divergences. The capacity is limited not because individual physicists lack intellectual seriousness but because the framework's epistemic status as the medium within which substantive empirical claims become formulable places the framework outside the coherence standard the discipline applies to its substantive predictions. This is the structural situation the essay has named. What remains is to address what the philosophy of physics, as a discipline whose subject matter is what physics does and how it does it, might productively do in response.

The essay's recommendation is that the philosophy of physics must take seriously the question of what alternative mathematical frameworks would look like, what their foundational commitments would be, and what physical theories formulated within them would predict. This recommendation is intentionally limited in scope. It does not propose that any specific alternative framework should be adopted by the discipline. It does not propose that the existing framework should be abandoned. It does not propose that physics is failing in any way correctable by an act of institutional will. The recommendation is that the alternative-framework question, currently treated as marginal or fringe by the mainstream of the discipline, must be elevated to a position commensurate with its structural importance.

The case for this elevation is not speculative. Work in the relational tradition has been proceeding for several decades, by working physicists from positions of disciplinary authority, with results that the broader discipline has not yet integrated into its self-understanding. The clearest contemporary statement of this work appears in Vidotto's recent essay on the relational ontology of contemporary physics.²⁹ Vidotto, working in quantum gravity at Western Ontario in collaboration with Rovelli, observes that relationality is not a marginal interpretation favored by a small subset of philosophers; it is the deep structure operative across the entirety of contemporary fundamental physics. Classical mechanics, when examined from the perspective of modern symmetry theory, treats physical properties as relations between systems rather than as intrinsic features of objects. Gauge theories treat gauge symmetries not as redundancies to be eliminated but as handles through which systems couple to one another, with the couplings themselves being

²⁹F. Vidotto, "The relational ontology of contemporary physics," in *Quantum Mechanics and Fundamentality: Naturalizing Quantum Theory between Scientific Realism and Ontological Indeterminacy*, edited by V. Allori (Springer, Cham, 2022), arXiv:2201.00907 [physics.hist-ph]. The diagnostic that relationality is the deep structure operative across contemporary fundamental physics is developed across the essay; the specific formulation that "the relational realism is not in any form relativist" appears at the close of the essay's concluding section.

the physical content. General relativity treats spacetime as a manifestation of the gravitational field's relational structure rather than as a fixed background within which dynamics occurs. Quantum field theory's notion of particle is itself relational, with the number of particles in a given state depending on the reference system that measures them. Quantum gravity, in its boundary-formulation, treats a spacetime region as a process and a state as the description of how two such processes interact. The relational structure is everywhere, in Vidotto's reading, and the discipline's framework of substance-based ontology is increasingly at odds with what its own theories are telling it.

What Vidotto's analysis does, with particular precision, is identify the strength of the relational reading without overclaiming for it. The reading is realist; it does not weaken the reality of the world. Going relational, in her formulation, does not collapse physics into subjectivism or perspectivalism in any sense that working physicists would find objectionable. The relations are among physical systems in interaction, not between observers and an unknowable substrate; conscious agents are a peculiar case among the different systems, not the central category. Information should be understood as physical correlations, without a necessary epistemic connotation. The relational ontology preserves objectivity by making it perspectival in the technical sense—multiple legitimate perspectives on the same physical reality, with no perspective enjoying ontological priority—while preserving the reality of what is perspectivally accessed. Her formulation, “the relational realism, it should be stressed, is not in any form relativist,” is precisely the formulation the essay's argument needs at this point, because it forecloses the most likely defensive objection a skeptical reader could raise. An alternative-framework conversation does not require abandoning realism. It requires reconceiving what the locus of the real is. The reconception is well within the philosophical resources the discipline already has.

The active research program in this direction is substantial. Rovelli's relational quantum mechanics, originally formulated in the 1996 paper that has become the standard reference, has been extended in subsequent work that addresses specific empirical puzzles within the relational reading. Smerlak and Rovelli's analysis of the EPR situation in 2007 demonstrated that the relational interpretation handles the entanglement paradox without requiring nonlocal action at a distance; the apparent paradox dissolves once the relativity of facts to interacting systems is taken

seriously.³⁰ Martin-Dussaud, Rovelli, and Zalamea's 2019 paper on the notion of locality in relational quantum mechanics extended this further, articulating how locality is preserved as a structural feature of the framework rather than as a contingent empirical fact requiring separate explanation.³¹ Together with Vidotto's broader synthesis, this cluster of work constitutes a research program—multiple authors, across multiple papers, across more than two decades, addressing specific physical and philosophical puzzles within a coherent alternative ontology. The cluster cannot be dismissed as marginal; it is sustained scholarship by physicists in good disciplinary standing, published in journals the field reads.

Vidotto's diagnostic work does not, however, complete the alternative-framework conversation. Her contribution is to identify that relationality is the deep structure operative across contemporary physics. What she does not provide—and what her work does not attempt to provide—is the foundational axiomatic framework in which this relationality would be made primitive rather than derived as a feature observed across existing sub-disciplines. The relational ontology Vidotto documents is, on her presentation, already there in physics, awaiting recognition. The further step of constructing a foundational framework in which interaction is taken as the sole primitive, with all other physical categories derived from it, is work that her analysis identifies as needed but does not itself undertake. Several proposals exist for what such a foundational framework might look like.

Constructor theory, developed by Deutsch and Marletto, attempts to reformulate physical laws as principles governing what physical transformations are possible and impossible, with the laws themselves understood as constraints on the constructors that perform the transformations.³² The framework abandons the standard dynamical-prediction format of physical law in favor of a counterfactual-structural format that, the authors argue, is more fundamental. Ontic structural realism, developed by Ladyman, Ross, French, and others, holds that the structural features of

³⁰M. Smerlak and C. Rovelli, "Relational EPR," *Foundations of Physics* 37, 427 (2007).

³¹P. Martin-Dussaud, C. Rovelli, and F. Zalamea, "The notion of locality in relational quantum mechanics," *Foundations of Physics* 49, 96 (2019).

³²D. Deutsch and C. Marletto, "Constructor theory of information," *Proceedings of the Royal Society A* 471, 20140540 (2015); C. Marletto, *The Science of Can and Can't: A Physicist's Journey through the Land of Counterfactuals* (Viking, New York, 2021).

reality are what physical theories are committed to, with the question of what underlies the structure deferred or denied as a category mistake.³³ The structural-realist position is broadly compatible with the relational reading Vidotto documents but emphasizes the philosophical analysis of what structure-as-fundamental means rather than the construction of a specific axiomatic framework. The Interactionist Mathematical Paradigm, developed in a recent monograph, proposes a specific axiomatic framework with interaction as the sole foundational primitive, with empirical applications across particle physics, cosmology, and the foundational philosophical puzzles physics inherits from its substance-ontological framing.³⁴

These alternative-framework proposals differ from one another in significant respects. Constructor theory is concerned with the structure of physical law; ontic structural realism is concerned with the philosophical interpretation of structure; the Interactionist Mathematical Paradigm is concerned with axiomatic foundations. The proposals are not interchangeable, and the philosophy of physics has not adjudicated between them. What they share is the structural commitment that the philosophical situation the essay has documented requires alternative-framework work, that such work is intellectually serious, and that its further development is the appropriate response to the structural diagnostic Vidotto and the present essay have separately articulated. The proposals are different answers to the question the essay's argument makes pressing. Whether any of them ultimately succeeds is a question the discipline must work out across the years and decades during which alternative-framework inquiry can continue. That the inquiry must continue, and must be taken seriously as continuation rather than as a marginal sideline, is what the essay's structural diagnostic establishes.

It is worth being honest about what the call to alternative-framework inquiry does and does not require. The alternative-framework programs cited above have been developed by working philosophers and physicists across multiple decades; in this sense, the inquiry the essay calls for is not work that needs to be initiated but work that needs to be recognized. What has been less

³³J. Ladyman and D. Ross, *Every Thing Must Go: Metaphysics Naturalized* (Oxford University Press, Oxford, 2007); S. French, *The Structure of the World: Metaphysics and Representation* (Oxford University Press, Oxford, 2014).

³⁴E. Arete, *Interactionist Mathematical Paradigm (IMP): A Formal Framework for Relational Ontology* (Zenodo, 2026), <https://doi.org/10.5281/zenodo.20486260>.

developed is the structural diagnostic that establishes why such work is necessary rather than merely interesting—why alternative-framework construction is a structural requirement for the discipline's empirical accountability at the foundational level, rather than an optional philosophical hobby pursued by those with the temperament for it. The present essay's contribution is to that diagnostic. The work has been done. What has been missing is the framework within which the work's necessity can be recognized. Providing that framework is the essay's purpose; recognizing the work that has already been done within it is the discipline's next step.

The call to alternative-framework inquiry raises an objection that the essay must address directly. If the existing mathematical apparatus is exempted from empirical accountability in the manner the essay has documented, by what criterion is an alternative framework to be evaluated? The existing framework was constructed to fit the empirical record; any alternative that achieves coherence with the existing record will be evaluable against the same standard the existing framework already meets, and the existing framework will, in many cases, win such comparisons because it was constructed to win them. Other criteria—elegance, parsimony, conceptual clarity—are available but are not themselves empirical criteria; appealing to them seems to abandon the empirical-accountability standard that motivated the call to alternatives. The objection is that the call to alternatives is bootstrapping: it depends on a criterion of evaluation that either reduces to the existing framework's self-vindication or escapes empirical accountability altogether.

The objection is real and deserves a direct answer. The criterion by which alternative frameworks should be evaluated is the same criterion the existing framework is supposed to meet, applied at framework level rather than at substantive-theory level: the Lakatosian progressiveness criterion. An alternative framework has legitimate claim on the discipline's attention if it can demonstrate three properties. First, empirical equivalence in established regimes—the alternative reproduces the existing framework's successes wherever the existing framework operates well. Second, superior handling of divergent cases—the alternative addresses the cases where the existing framework requires elaborate accommodation, doing so with greater parametric economy or structural simplicity. Third, novel predictions—the alternative generates empirically distinguishable predictions in regimes where no current framework has been tested. These three criteria are not circular because they are not asking the alternative to justify itself by the existing framework's standards. They are asking the alternative to meet the same standard the existing

framework was supposed to meet, applied at framework level rather than only at substantive-theory level. The difference is that the standard is now being applied to the framework itself rather than only to the theories built within it.

The progressiveness criterion is necessary but not sufficient for the diagnostic the essay has developed. What the diagnostic specifically requires is alternative frameworks whose practitioners commit explicitly to the framework's *ansatz* status—to its standing as a working hypothesis subject to revision in light of evidence, not as an established starting point exempt from further empirical challenge. This commitment is structural rather than mathematical; the same mathematical apparatus, held differently, becomes structurally accountable in ways it currently is not. An alternative framework that is empirically progressive in the Lakatosian sense but is held by its practitioners as a new fixed substrate would, over time, reproduce the structural situation the essay has documented in the current framework. The diagnostic is not solved by replacing one unrevisable substrate with another; it is solved by adopting a framework whose unrevisability is itself revised—by communities of practitioners who explicitly treat the framework as an *ansatz* that the empirical record may eventually require them to abandon. *Ansatz* status is not a confession of weakness. It is the acknowledgment of empirical accountability that the essay's diagnostic identifies as currently absent at the framework level. Without it, even a progressive alternative becomes the next unfalsifiable substrate. With it, the framework is structurally accountable in the dimension the diagnostic identifies as currently exempted.

The criteria as stated leave open a question that should be addressed explicitly: multiple alternative frameworks may satisfy both progressiveness and *ansatz* commitment without uniquely identifying which alternative the discipline should adopt. This is a feature rather than a defect, and it lies along a spectrum of difference worth naming. At one end of the spectrum, alternative frameworks may differ only at the level of notation and convention while sharing identical ontological content—as Newton's fluxional notation and Leibniz's differential notation differed in convention while expressing the same calculus. At the other end, alternative frameworks may differ at the level of foundational formal apparatus while ultimately proving empirically equivalent—as Heisenberg's matrix mechanics and Schrödinger's wave mechanics differed in their mathematical structures and intuitive pictures while being shown by Dirac and von Neumann to be equivalent formulations of the same theory. Between these endpoints, alternatives can differ

at varying depths of structural commitment, with the empirical equivalence being more or less complete. The structural-preservation diagnostic the essay has developed applies most pressingly to frameworks at the Heisenberg-Schrödinger end of this spectrum and beyond, where the formal apparatus carries different ontological commitments. Frameworks at the Newton-Leibniz end do not really differ in their commitments; they differ in their expression of identical commitments. The discipline's eventual adoption of Leibniz's notation over Newton's was driven by considerations of calculational convenience and pedagogical clarity, not by empirical superiority—and this is a legitimate basis for framework choice when the frameworks are empirically equivalent. Multiple correct alternatives may coexist; the discipline's task is not to identify the uniquely correct one but to develop alternatives to the point where progressiveness comparison becomes possible, with the choice among empirically equivalent and structurally adequate alternatives being a secondary question to be settled on grounds the philosophy of physics has historically recognized as legitimate.

The closing observation is that the discipline's empirical accountability at the foundational level depends on the inquiry being undertaken. A framework that is not subject to coherence with evidence at its foundational level can only be rendered accountable by the construction and serious evaluation of alternatives. Without alternatives, the framework's relationship to evidence is asymmetric in the way the essay has documented; with alternatives developed to a comparable level of formal sophistication and empirical applicability, the framework can be evaluated against them on coherence grounds, and the structural exemption identified in the previous sections can be addressed. The alternative-framework inquiry is therefore not a competitor to the existing framework; it is the procedure through which the existing framework can become accountable in the dimension where it is currently not. This is the strongest reason the inquiry must proceed. It is also the reason the philosophy of physics, as the discipline positioned to take the inquiry seriously, has a structural responsibility to do so. The inquiry will not be performed by physics, because physics does not have the procedure to perform it on itself. It will be performed by the philosophy of physics, if it is performed at all, and the responsibility of the discipline is to recognize that this is its work.

The essay closes here. The argument has been that physics, despite its self-presentation as the paradigmatic empirical science, operates as a two-layer system in which the substantive theory

layer is subject to coherence with evidence while the foundational mathematical apparatus layer is subject only to internal consistency. The asymmetry has consequences that the essay has documented across five cases. The cases collectively demonstrate that the framework-preservation dynamic operates as a structural feature of contemporary physics, not as a series of unrelated puzzles. The structural feature has costs the discipline has not adequately reckoned with, including the accumulation of unfalsifiable elaborations, the foreclosure of alternative-framework inquiry on coherence grounds, and the gradual disconnection of the framework from the empirical accountability the discipline claims for itself at the foundational level. The response the essay recommends is that the philosophy of physics take seriously the existing alternative-framework research, of which the relational tradition documented by Vidotto and the foundational proposals in constructor theory, ontic structural realism, and the Interactionist Mathematical Paradigm are present examples—and that such alternatives be evaluated by the criteria the essay has specified: Lakatosian progressiveness applied at framework level, combined with explicit ansatz commitment. The work is being done. The discipline's task is to recognize that this work is what addresses what the essay has named: the unfalsifiable substrate of empirical physics.